

# Capstone Project Introduction

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ECON 490

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# Overview

In these slides, we'll talk about:

- Requirements for the capstone project
- Basic components of your projects
- Tips for your proposals and common data analysis pitfalls

# Key Components of Your Capstone Papers

1. Clear research question and motivation
2. Data analysis

We'll complete several intermediate assignments throughout the semester

- **Research Proposal:** Starting point for your projects
- **Week 8 CPS Lab (in-class activity):** Learn how to produce data analysis output
- **Capstone Data Lab:** Repeat same pipeline using your own capstone data
- **Capstone Outline:** Will include draft data analysis and short presentations

Proposal/Outline assignments have detailed templates for you to use/submit

# Capstone Project Grade Breakdown

The capstone project is worth **40%** of your final grade

| <i>Assignment</i>    | <i>Due Date</i>          | <i>% of Final Grade</i> |
|----------------------|--------------------------|-------------------------|
| Research Proposal    | Friday, 10/9             | 5%                      |
| Capstone Data Lab    | Wednesday, 10/21         | 7.5%                    |
| Outline Presentation | Wednesday, 11/4          | 2.5%                    |
| Outline              | Friday, 11/6             | 7.5%                    |
| Final Presentation   | Weeks 14/15 (12/2, 12/9) | 7.5%                    |
| Final Paper          | Friday, 12/18            | 10%                     |

# Final Deliverables

All capstone output should be:

- Written in your own words
- Neatly formatted/presentable (no R screenshots unless specifically requested)

**Final Papers:** Roughly 8-10 pages

- Including data analysis output and works cited
- We will discuss writing and paper requirements later in the semester

**Presentations**

- *Final Presentations:* ~15 minutes
- *Outline Presentations:* 5-10 minutes

# Capstone Research Proposal

First capstone assignment is due Week 7 (Friday, 10/9)

Steps to complete this assignment:

- 1) Decide if you want to work alone or with a partner
- 2) Identify a clear research question
- 3) Identify the data you'll use to answer your research question
- 4) Complete the proposal worksheet on Canvas

***NOTE: Before moving forward with your project, you will need approval on the topic and data source(s) – I may ask you to submit a revised proposal.***

# Working Backwards

Trying to find a research question can be daunting

To make things easier, try *working backwards*

- Start thinking about the general topics you're interested in – this could be labor markets, housing, education, poverty, etc.
- Then ask, “What kind of data analysis can I do?” related to that topic

Your data analysis requires (1) data and (2) an analysis strategy

- Start by (1) identifying and exploring the data set that best fits your interest
- Then, think about (2) possible analysis strategies
- Finally, return to your research question and orient around your analysis

# General Suggestions on Proposals

Key first step: find and download data (ideally multiple data sets)

- Load data in R, read documentation (!), run simple regressions
- Ask yourself, “Does this data look like something I can work with?”

Your first draft of a research question will likely be too superficial

- Remember, the goal is to write an 8+ page paper
- As part of this, you’ll spend 2-3 pages talking about your data analysis output

*If your research question is something like, “What’s the correlation between GDP and unemployment?” you’re not going to have much to say!*

# Selection of Research Questions from Prior Semesters

*How do recessions impact household food expenditures?*

*How do the returns to graduate education vary across demographic groups?*

*How did the COVID-19 pandemic impact household transportation choices?*

*What factors drive variation in home prices across counties in California?*

*How much of NBA player compensation can be explained by future expected performance (compared to past performance or prior accolades)?*

# Thinking Descriptively

How to go beyond, “What’s the correlation between X and Y?”

- For ECON 340, running 1-2 regressions is generally sufficient for a paper
- In 490, goal is to concretely answer a question/say something interesting

In general, think about *exploring variation* in either:

1. The level of one economic variable as a function of other variables
2. A relationship between economic variables

In both cases, you can explore variation across time, places, economic factors, etc.

# Two Components for Your Capstone Data Analysis

As part of your capstone paper, you'll include the following output:

**Summary Output:** *What does your data look like?*

- Summary stats table for outcome + explanatory variable(s)
- At least 2 graphs or maps for your outcome + explanatory variable(s)

**Results Output:** *Results in (at least) two stages*

- 1<sup>st</sup> stage is a “first pass” answer to your research question
- 2<sup>nd</sup> stage builds on and develops the 1<sup>st</sup> stage along some dimension
- For most projects, this entails 2+ regression tables (some projects may differ)

# Common Question: “*Is Just Doing X Good Enough?*”

**Goal for Project:** Be able to say/show something interesting

- This means I can't have a cookie-cutter answer to what's good enough

**General Rule of Thumb:** Every project should entail roughly the same amount of work

- If you downloaded a single data set from a website (IPUMS, etc.), your analysis output should be relatively clean and thorough
- If you collected a lot of data manually/by hand, I take that into account

**Rough Guideline:** You should have more than 1 piece of output for at least 1 stage of your results (i.e., multiple regression tables/figures in one of your stages)

## Common Question: “What if I add an $X$ variable (or swap $Y$ )? Is that a new stage?”

Suppose you start by running  $\text{wage} \sim \text{education}$  using ACS data

- Then, you run a second regression:  $\text{wage} \sim \text{education} + \text{age}$
- Age is correlated with  $Y$  &  $X$  so it's an OV—why not just include it in baseline?

In general, you should include ***all*** reasonably available explanatory variables in 1<sup>st</sup> stage, then run something substantively different in 2<sup>nd</sup> stage

- *Caveat*: If you do a lot of manual data collection, this requirement is relaxed

# Data Analysis Pitfalls: *Macro Time Series Data*

**General Rule of Thumb:** Make sure you have data for multiple groups

- In a lot of data sets/research contexts, this happens naturally
- People in different states, players on different NBA teams, etc.

**SPECIFIC GUIDELINE:** Avoid ideas using data on a single place/company/etc. over time

- Lack of groups = lack of natural comparisons to make via regression
- Severely limits flexibility in your analysis = hard to write an 8-10 page paper

Want to do a macro topic? Look up data for multiple countries, or states in US

# Data Analysis Pitfalls: *Ecological Fallacy*

A common 490 topic that I've decided to prohibit: "*What is the correlation between state economic conditions and crime?*"

Two problems:

- 1) Correlations are generally a bit weak
- 2) Correlations are very easy to misinterpret

**Ecological Fallacy:** Making inferences about individuals based on group-level data

- Asking, "How does losing *my* job affect *my* propensity to commit crime?" is very different from asking, "How do recessions impact crime *in general*?"

# Data Analysis Pitfalls: *Getting Too Fancy*

In general, OLS should be fully sufficient for most projects

- If you have a background/prior course experience with ML/time-series econometrics/etc., you are welcome to apply these tools!
- However, don't let AI push you into using techniques you haven't studied

Using a fancy method doesn't make analysis better if the question isn't well phrased or the data isn't well-suited to your analysis

ChatGPT loves logit/probit regression and variance inflation factors

- Avoid these tools unless you 1) know why you're using them and 2) know how to correctly interpret their output

# Miscellaneous Suggestions

**Key Advice:** More time spent collecting/cleaning data now means an easier project going forward—give yourself time to adjust!

You are welcome/encouraged to adjust your data & specific research question

- **Key Requirement:** email me if you substantively change either your data or your overall research focus

If your project requires collecting/processing 2+ data sets (that aren't clean/easy to merge), your stages can be distinct parts of your analysis

- Think “two parallel stages of analysis” instead of sequential stages